Seyed Milad Hatamlee, PhD

RESEARCH INTERESTS

Molecular simulation and statistical thermodynamics of confined fluids; nanoscale thermal transport and interfacial phenomena; atomistic mechanics and defect engineering in low-dimensional materials; regression-based machine learning for structure–property modeling; thermodynamic model calibration; adsorption and carbon-management materials.

EDUCATION

PhD in Mechanical Engineering, GPA: 3.94/4.0 Jan. 2022 – May 2026
University at Buffalo, State University of New York, Buffalo, NY
Dissertation: Thermodynamics and Transport of Small Molecules under Nanoconfinement
Advisor: Dr. Jiaoyan Li

MS in Mechanical Engineering (Energy Conversion), GPA: 3.58/4.0 Sep. 2017 – Feb. 2020
Imam Khomeini International University (IKIU), Qazvin, Iran
Thesis: Heat Transport in 2D Structures of Carbon Nitride by Non-Equilibrium Molecular Dynamics
Advisor: Dr. Ali Rajabpour

BS in Mechanical Engineering Sep. 2013 – Sep. 2017
K.N. Toosi University of Technology (KNTU), Tehran, Iran
Thesis: Comparison of Ventilation Systems for Large Commercial Projects from an Economic Perspective
Advisor: Dr. Ali Behbahani Nia

RESEARCH EXPERIENCE

Research Assistant — University at Buffalo Jan. 2022 – May 2026
Multiscale/Multiphysics Theory and Simulation (MulTS) Laboratory

- Built Python and Bash workflows supporting high throughput confined-fluid simulations, automating HPC job submission, output extraction, post-processing, and visualization.
- Applied GEMC and gauge-GCMC simulations to characterize vapor–liquid equilibrium of CO_2/n-heptane and methane/n-heptane mixtures under bulk conditions and confinement in hydroxylated silica pores.
- Calibrated attraction and covolume parameters in a modified Peng–Robinson equation of state using genetic optimization, reducing normalized RMSE against Monte Carlo phase-equilibrium data by 27–61% relative to the baseline confined model at the fitted conditions.
- Modeled pressure-driven water, CO_2, and dilute water–CO_2 transport in 4–8 nm silica nanochannels, quantifying how surface hydroxylation and graphene coatings affect velocity, interfacial slip, and apparent permeability.
- Analyzed molecular density profiles and hydrogen-bond populations to explain surface-dependent flow enhancement and suppression relative to continuum predictions.
- Investigated competitive water/CO_2 adsorption and transport in zeolite 13X, linking humidity-dependent CO_2 capacity loss to occupation of adsorption sites by water.

Research Assistant — IKIU Feb. 2018 – Nov. 2021
Advanced Simulation and Computing Laboratory (ASCL)

- Calculated thermal conductivity across 15 carbon-based two-dimensional materials using LAMMPS NEMD, assessing directional and finite-size effects and extrapolating conductivity to infinite length.
- Trained regression models relating thermal conductivity separately to density and Young's modulus across a small simulation dataset, establishing interpretable structure–property relationships.
- Combined NEMD, GULP lattice dynamics, and Python-based vibrational-mode classification to investigate 15 material interfaces; fitted power-law relationships between localized interfacial-mode populations and thermal conductance.
- Quantified how point vacancies and line defects affect phosphorene's tensile and shear response, including elastic moduli, fracture strain, failure initiation, and mechanical anisotropy
- Compared stress–strain response and fracture properties across eight titanium carbide/nitride MXenes, evaluating oxygen termination and species-specific vacancies in Ti_2N.

Selected Collaborative Research

- Investigated thermal conductance across gold–silica interfaces using NEMD, assessing crystalline versus amorphous silica, temperature, and interaction strength through vibrational analysis and comparison with a diffuse mismatch model.
- Modeled dry and hydrated calcite–mineral interfaces to quantify binding energies and adhesion, helping explain experimental strength differences in biocemented sands.

Selected Finite-Element Projects

- Applied Abaqus finite-element analysis to stress concentration, crack propagation, thermal expansion, and contact problems.
- Analyzed stress–strain response and deformation under prescribed loading and boundary conditions, interpreting results using mechanics and materials principles.

TEACHING EXPERIENCE

Course Instructor — MAE204 Thermodynamics I Summer 2024

Department of Mechanical and Aerospace Engineering, University at Buffalo

- Independently designed course assessments and recitation sessions for 40+ undergraduate students.

Teaching Assistant — MAE385 Engineering Materials Lab Spring 2023, 2024, 2025, 2026

Department of Mechanical and Aerospace Engineering, University at Buffalo

- Led 5 independent lab sections per semester across 4 years, guiding 200+ students per semester in hands-on materials characterization, mechanical testing, and microstructural analysis.

Teaching Assistant — MAE381 Engineering Materials Fall 2022, 2023, 2024, 2025

Department of Mechanical and Aerospace Engineering, University at Buffalo

- Supported 300 students per semester through grading, office hours, and clarification of complex materials science concepts including crystal structure, defects, phase diagrams, and mechanical behavior.

PUBLICATIONS

1. Hatamlee SM, Hoepfner M, Elmehlawy M, Lu M, Li Z, Li J. Modified Peng–Robinson equation of state for CO_2/hydrocarbon binaries under confinement based on Monte Carlo simulations. Accepted at Langmuir, 2026.
2. Hatamlee SM, Li J. Molecular dynamics study of water flow in silica nanochannels: transition from flow enhancement to reduction. Journal of Molecular Liquids, 2026 Jul 13:129795.
3. Xia Y, Liu J, Kancharla R, Li J, Hatamlee SM, Ren G, Semeykina V, Hamed A, Kane JJ. Insights into the 3D permeable pore structure within novel monodisperse mesoporous silica nanoparticles by cryogenic electron tomography. Nanoscale Advances, 5(11):2879–2886, 2023.
4. Ghasemi H, Hatamlee SM, Tirkolaei HK, Yazdani H. Biocementation of soils via enzyme induced carbonate precipitation: An integrated laboratory and molecular dynamics study. Biophysical Chemistry, 284:106793, 2022.
5. Hatamlee SM, Jabbari F, Rajabpour A. Interfacial thermal conductance between gold and SiO2: A molecular dynamics study. Nanoscale and Microscale Thermophysical Engineering, 2022.
6. Hatamlee SM, Gordiz K, Rajabpour A. Lattice dynamics based descriptors for the interfacial heat transfer across 2D carbon-based nanostructures. Journal of Applied Physics, 2021.
7. Arabha S, Aghbolagh ZS, Ghorbani K, Hatamlee SM, Rajabpour A. Recent advances in lattice thermal conductivity calculation using machine-learning interatomic potentials. Journal of Applied Physics, 130(21):210903, 2021.
8. Hatamlee SM, Esfandiar A, Rajabpour A. Mechanical behaviors of titanium nitride and carbide MXenes: A molecular dynamics study. Applied Surface Science, 2021.
9. Hatamlee SM, Peer-Mohammadi H, Rajabpour A. Tuning shear mechanical properties and tensile strength anisotropy of monolayer black phosphorene: A molecular dynamics study. Materials Today Communications, 2020
10. Hatamlee SM, Rajabpour A, Volz S. Thermal conductivity of graphene polymorphs and compounds: from C3N to graphdiyne lattices. Carbon, 161:816–826, 2020.

CONFERENCE PRESENTATIONS

- Hatamlee SM, Li J. Molecular dynamics study of water flow in silica nanochannels: surface wettability governs the transition from reduction to enhancement. American Physical Society, Division of Fluid Dynamics 78th Annual Meeting, Nov. 2025.
- Hatamlee SM, Li J. Modified Peng–Robinson equation of state for CO_2/hydrocarbon binaries under confinement based on Monte Carlo simulations. MUSE All-Hands Meeting, University of Utah, Salt Lake City, Jan. 2024.
- Hatamlee SM, Rajabpour A, Volz S. Thermal conductivity of graphene polymorphs and compounds: from C3N to graphdiyne lattices. 7th National Conference on Nanotechnology from Theory to Application, Jun. 2019.

TECHNICAL SKILLS

Atomistic Modeling: Molecular dynamics (LAMMPS); Monte Carlo, GEMC, GCMC, and gauge-GCMC (Towhee); DFT (VASP, Quantum ESPRESSO); lattice dynamics (GULP)
Materials Modeling: Thermodynamics, phase equilibria, thermal transport, defect effects, mechanical properties, model calibration, sensitivity analysis

Continuum Modeling: Abaqus finite-element analysis; stress concentration, crack propagation, thermal expansion, contact; Continuum transport analysis

Machine Learning and Statistical Modeling: Regression-based property modeling, nonlinear regression, structure–property descriptors, scikit-learn, model-fit assessment using MAE and RMSE
Scientific Programming: Python (NumPy, SciPy, pandas, Matplotlib), MATLAB, Bash

Computing and Analysis: High-performance computing, Linux, Git, automated simulation workflows, OVITO, data visualization

AWARDS, HONORS & CERTIFICATES

2025 Graduate Scholarship — J. & S. Niciszewska Mucha Endowed Scholarship, University at Buffalo ($10,000)
2024 Leadership Grant — Council of Advocacy and Leadership Programming, UB ($6,500)
2013 Ranked top 738 among 250,000+ participants, Iranian University Entrance Exam
2019 Certificate: A Not-so-brief Introduction to Molecular Biophysics, IPM, Tehran
2018 Certificate: Quantum Calculation using Quantum ESPRESSO, Sharif University of Technology
2018 Certificate: 1st Conference on the Computation of Nanostructures, Sharif University of Technology

SERVICE TO PROFESSION

Leadership	2024–2025	President, MAE Graduate Student Association, University at Buffalo
	2022–2024	President, Iranian Graduate Student Association (IGSA), University at Buffalo
Peer Review	2022, 2024	Journal of Applied Physics